SHAHSAWAR ORAKZAI

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer specializing in end-to-end production systems, from **RAG architectures** and autonomous **AI Agents** to **WebGL** frontends and containerized backends. Strong track record of architecting, deploying, and maintaining high-performance applications independently in remote, async environments.

EXPERIENCE

Freelancer.com — Feb 2026 – Present
Full-Stack AI Engineer (Self-Employed) — Remote

- Delivered enterprise **RAG assistants** and document retrieval systems for business clients with **FastAPI/pgvector** backends and responsive **TypeScript** frontends, cutting manual inquiry response times by **~65%**.
- Orchestrated autonomous **agentic workflows** using Hermes and function-calling architectures with persistent memory and external tool integration, automating multi-step operational processes.

MindGigs — Dec 2025 – Feb 2026
Python Developer Intern — Peshawar, Pakistan

- Engineered high-throughput **REST APIs** using **FastAPI** and **asyncio**, boosting request throughput by **~40%** with strict **Pydantic** schema validation.
- Integrated local **Small Language Models** (SLMs) and developed **Dockerized** data scraping pipelines processing **20k+ daily records**.

Pakistan Swimming Federation — May 2024 – Sep 2025
Technical Operations Lead (Part-Time) — Peshawar, Pakistan

- Designed and deployed a centralized management information system for **200+ registered athletes**, replacing paper-based records with a searchable database.
- Served as technical lead managing official entries for **30+ provincial swimmers** across district and national championships, automating administrative workflows using **Python** and Excel.

PROJECTS

3D Portfolio with RAG Chatbot — BOTRON (Oct 2026) — Three.js, WebGL, FastAPI, Groq, Supabase, GitHub Actions
Live Site • Source Code

- Built a scroll-driven **WebGL** portfolio featuring a 3D robot with **inverse-kinematics** gaze tracking and adaptive frame rates (60/30/0 FPS), deployed on **GitHub Pages**. Pre-launch content reached **50,000+ impressions** and **500+ likes** on LinkedIn.
- Deployed **BOTRON**, a full **RAG** chatbot with **FastEmbed** vector search over 16 knowledge chunks, a minimum-similarity floor to prevent hallucinations, and 3-model **Groq** failover with backend deployed on **FastAPI Cloud**.
- Connected **Supabase** to capture and analyze all user queries, enabling data-driven content improvements across the portfolio.

Rednote — Document Retrieval Assistant (Jun 2026) — Python, pgvector, Neon PostgreSQL, TypeScript, Docker, CI/CD
Live App • Source Code

- Developed a full-stack document retrieval assistant with **LLM-powered search** and **pgvector** semantic indexing, achieving **sub-250ms query response times** across **10,000+ indexed chunks**.
- Engineered an automated document ingestion pipeline with **JWT** authentication and guest-to-registered conversion, onboarding **30+ active users**.
- Containerized application with **Docker** and deployed to production via automated **GitHub Actions CI/CD** pipeline, reducing deployment cycles to under 3 minutes.

Async Web Scraper (Apr 2026) — Python, curl_cffi, Neon PostgreSQL, TypeScript, Groq LLM, Docker
Live App • Source Code

- Engineered an async scraping engine using **curl_cffi** with browser TLS impersonation to bypass Cloudflare bot detection, concurrently scraping **30+ raw web pages in parallel**.
- Architected an **8+ Groq LLM rotation pool** with adaptive rate-limiting and automatic failover, structuring raw HTML into downloadable **JSONL** datasets for model fine-tuning.
- Deployed live on **Hugging Face Spaces** backed by a serverless **Neon PostgreSQL** database, featuring a **TypeScript** UI with **3s interval polling** that cut manual verification overhead by **90%**.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, C++, SQL (PostgreSQL), Bash
AI / LLM	RAG Architectures, AI Agents, LangChain, LangSmith, pgvector, FastEmbed, Groq, OpenAI, Open-Source LLMs
Backend	FastAPI, Pydantic, SQLAlchemy, AsyncIO, Supabase, Neon PostgreSQL, REST APIs, WebSockets
Frontend & 3D	Three.js, WebGL, GSAP, HTML5, CSS3, React, Vite
DevOps & Tools	Docker, Git, GitHub Actions, CI/CD, FastAPI Cloud, n8n Automation, Linux, Hugging Face

EDUCATION

University of Agriculture, Peshawar — Oct 2024 – Present
Bachelor of Science in Computer Science (BSCS) — CGPA: 4.0

CERTIFICATIONS & ACHIEVEMENTS

CS50X: Introduction to Computer Science (Harvard University) Verify — Feb 2026
CS50P: Introduction to Programming with Python (Harvard University) Verify — Oct 2025

National Athlete: Represented KP province for 10 years as a National Swimmer, winning 10+ medals including Gold at National Junior Championships.